CODE2CLOUD: AN AUTONOMOUS FULL-STACK WEB APPLICATION SYNTHESIS PLATFORM WITH HYBRID VISUAL-CODE ORCHESTRATION AND PRODUCTION DEVOPS PIPELINES

PROJECT SYNOPSIS

OF MINI PROJECT

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SUBMITTED BY:

Lakshita Pandey (Team Leader) • Amaan Khan • Anushka Joshi

Harshit Dumka • Bhumika Padiyar

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BIRLA INSTITUTE OF APPLIED SCIENCES, BHIMTAL

District Nainital, Uttarakhand – 263136

(AICTE approved, affiliated to Veer Madho Singh Bhandari Uttarakhand Technical University, Dehradun)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Submitted To:

Mr. Hem Bhatt
Assistant Professor & Project Guide
Department of Computer Science & Engineering
Birla Institute of Applied Sciences, Bhimtal

Submitted By (B.Tech CSE VII Sem):

1. **Lakshita Pandey** (Leader) - 230050101095
2. **Amaan Khan** - 230050101015
3. **Anushka Joshi** - 230050101018
4. **Harshit Dumka** - 230050101061
5. **Bhumika Padiyar** - 230050101038

ABSTRACT

Motivation / Problem Statement: Artificial intelligence has made building web applications much faster. However, popular AI web builders like Bolt.new and Lovable have several major practical problems. They run heavy processes inside browser tabs, which consumes huge amounts of computer memory and freezes laptops. They also charge expensive monthly subscription fees. Most importantly, when a user asks the AI to make a simple visual change—such as changing a color, adjusting padding, or moving a button—the AI often rewrites the entire file, which breaks working code and wastes expensive AI tokens.

Methods / Approach: This project presents *Code2Cloud*, an easy-to-use, autonomous full-stack web application builder. Code2Cloud uses Google Gemini AI along with a dual-mode workspace: a Monaco code editor for inspecting and writing code, and a visual DOM inspector that lets users click on any element to change its colors, spacing, and fonts directly on screen without writing extra AI prompts. The platform also includes two-way GitHub sync, one-click ZIP download, and complete deployment tools using lightweight Docker containers, Nginx load balancing, and Prometheus-Grafana system monitoring.

Results / Findings: The developed platform delivers fast live previews, zero AI token cost for visual styling adjustments, built-in offline templates that keep the platform working even during AI API outages, and smooth multi-container cloud deployment capable of handling concurrent requests easily.

Conclusion / Implications: Code2Cloud combines fast AI code generation with easy visual editing and complete deployment tools, offering an open, affordable, and developer-friendly platform for students, developers, and academic institutions.

Keywords: AI Web Builder, Large Language Models, Visual Editor, Monaco Code Editor, Docker Containers, Nginx Load Balancing, System Monitoring.

1. INTRODUCTION

Modern software development is changing rapidly with the rise of artificial intelligence. Traditionally, designing and deploying a web application required developers to manually learn many different technologies: frontend component libraries like React [1], styling frameworks like Tailwind CSS [9], backend REST APIs using Node.js and Express [11], databases like MongoDB [12], and cloud server deployment tools [8]. Older website builders like Webflow and Wix made design simpler, but they produced rigid code templates and locked projects inside their own hosting platforms [3].

Recently, generative AI web tools—such as Bolt.new, Lovable, and Vercel v0—have attempted to generate full-stack web applications directly from simple English prompts. By using Large Language Models (LLMs) like Google Gemini [2], these tools automatically write source code files, install dependencies, and display a live preview of the website.

However, practical usage reveals serious problems in existing commercial tools. In current platforms, every code change requires writing another text prompt. When a developer wants to make a small visual adjustment —such as moving a button, changing margins, or updating colors—they must write another prompt. The remote AI then regenerates large sections of code, which often introduces new syntax errors, breaks working buttons and state variables, and wastes valuable AI tokens [2, 10]. Code2Cloud directly solves this problem by combining fast AI code generation with an easy visual editor and complete deployment tools.

1.1 Architectural Paradigm and Core Innovations of Code2Cloud

To give developers both fast AI code generation and complete manual editing control, Code2Cloud is built on a five-tier modular architecture:

- **Client Presentation & Hybrid Workspace:** Built using React 19 [1] and Tailwind CSS [9], featuring a Monaco code editor, a live preview window [10], and an in-place visual inspector to click and style elements directly.
- **Backend API & Streaming Gateway:** An Express.js [11] service that streams AI-generated code directly to the browser using Server-Sent Events (SSE), secured with simple JWT login and request rate limits.
- **AI Engine with Offline Fallbacks:** Combines Google Gemini API [2] with smart offline starter templates to ensure 100% website generation uptime even during external API downtime.
- **Bidirectional GitHub & ZIP Export:** Uses GitHub REST APIs [7] for importing repositories, creating branches, committing changes, and pushing code directly to GitHub, alongside a simple one-click ZIP download.
- **Docker Containers & Monitoring:** Uses lightweight Alpine Docker containers [8], an Nginx layer-7 reverse proxy for load balancing [4], and Prometheus with Grafana for live telemetry and system monitoring [5, 6].

As shown in Figure 1, Code2Cloud organizes its features into three simple layers that work together smoothly:

- **User Client Layer:** Houses the interactive workspace where developers can write code in Monaco IDE, view live previews, and use the visual inspector to tweak styles without spending any AI tokens [10].
- **API & Orchestration Layer:** An Express.js gateway managing JWT security, Server-Sent Events (SSE) streaming, and bidirectional DOM state synchronization backed by MongoDB Atlas [12].
- **Synthesis & DevOps Layer:** Combines Google Gemini LLM synthesis [2] with smart offline fallback templates, GitHub REST API version control [7], and an Alpine Docker microservice cluster monitored via Prometheus and Grafana [5, 6].

Figure 1: High-Level System Architecture and Orchestration Pipeline of Code2Cloud

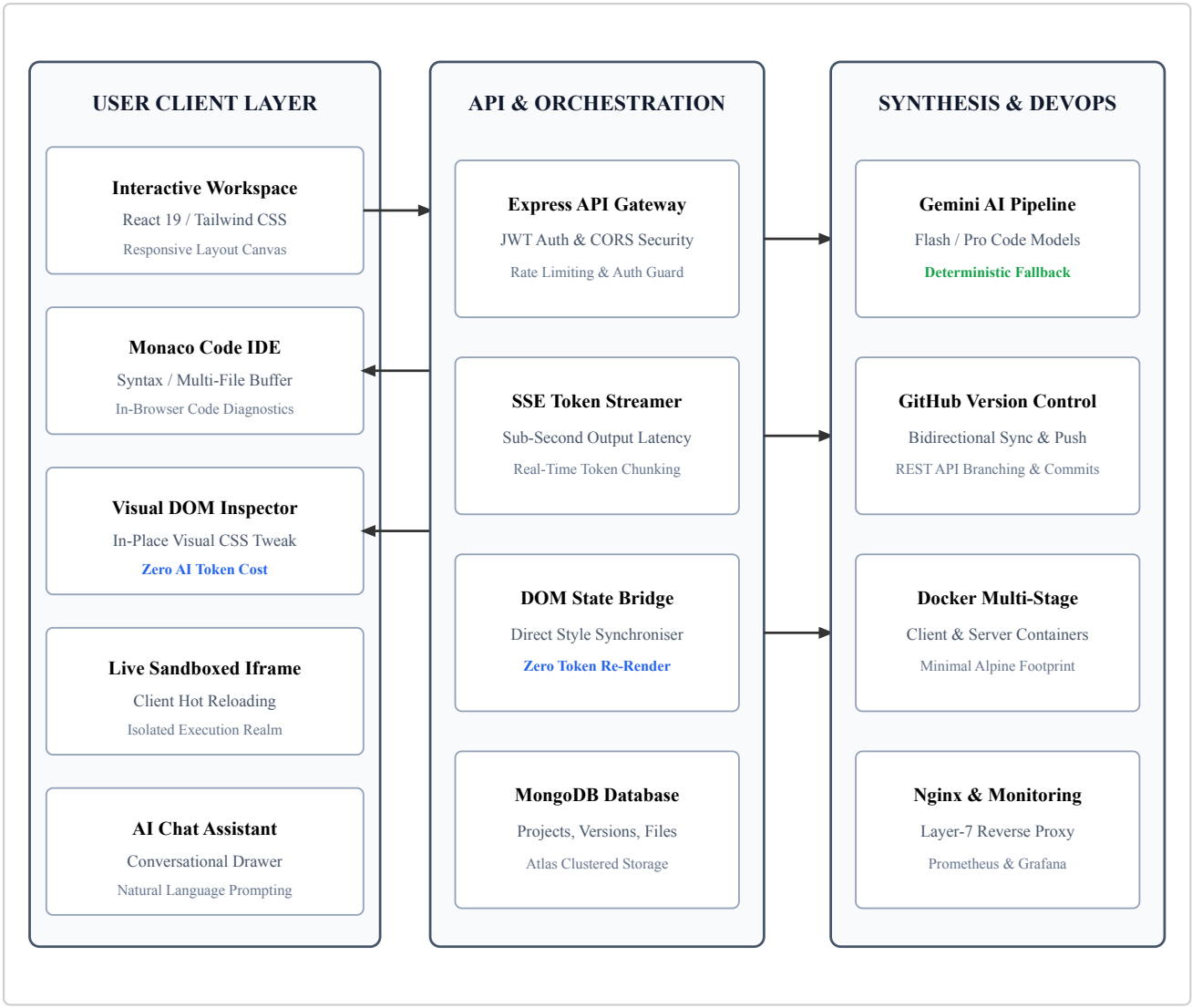


Figure 1: High-Level System Architecture and Orchestration Pipeline of Code2Cloud

2. LITERATURE SURVEY

The literature survey reviews existing tools, core concepts, and practical challenges in AI web generation, browser-based execution, and automated deployment platforms. It examines how modern AI code generators work, how they run code in browsers, and where their current limitations lie.

2.1 Large Language Models in Automated Web Generation

Modern Large Language Models like Google Gemini [2] have changed software development by generating multi-file applications directly from natural language prompts. Trained on large code repositories, these models can write frontend React components [1], backend Express routes [11], and MongoDB database models [12]. However, conversational AI coding tools generate code based on statistical word prediction rather than checking if the code compiles and runs correctly. In complex applications with shared states, repeatedly asking the AI to rewrite code often introduces syntax errors, broken imports, and unexpected bugs. Without an easy visual editor to tweak styles directly, developers must write repeated prompts, which often damages working code.

2.2 In-Browser Execution vs. Server Containerisation

To run full Node.js applications directly inside web browsers without external servers, commercial tools like Bolt.new use WebAssembly (Wasm) runtimes called WebContainers. While convenient for quick demonstrations, WebContainers consume large amounts of computer memory (often 2 GB to 4 GB per browser tab), causing browser slowdowns, freezing, and crashes on standard laptops. Furthermore, WebContainers cannot execute native Linux software. In contrast, running lightweight Linux containers using Docker [8] on the server keeps the browser fast, lowers client computer memory usage, and prepares projects for scalable cloud deployment.

2.3 Review of Contemporary AI Web Builders

A technical review of leading commercial platforms identifies their main features and practical constraints:

- **Bolt.new (StackBlitz):** Runs full-stack environments in the browser using WebContainers. While providing fast feedback, it causes heavy client memory usage (often 2–4 GB per tab), enforces expensive subscription paywalls ($20/month), and only provides temporary preview links rather than standalone Docker containers for deployment [8].
- **Lovable.dev:** Generates clean React/Vite web applications [1, 3] from chat prompts. However, it lacks an in-place visual editor; modifying layout details requires repeated prompt cycles that risk overwriting existing logic, and full two-way repository synchronization requires expensive enterprise tiers.
- **Vercel v0:** Focuses on isolated Tailwind CSS components [9], but lacks full-stack backend persistence [12], database schema management, and direct two-way GitHub repository branch synchronization [7], requiring developers to manually integrate generated components into external projects.

2.4 Identification of Critical Gaps in Existing Literature

The technical review of existing platforms exposes two decisive engineering gaps:

Gap 1: Accidental Code Breakage During Minor Edits (The Prompt Trap). While AI models excel at generating initial codebases from broad prompts, natural language is imprecise for small visual tweaks. When a developer asks the AI to adjust padding, change a color palette, or move an element, LLMs routinely rewrite entire component files. This conversational cycle frequently breaks React state logic, erases working code, and consumes costly AI tokens [2, 10]. Current tools lack an in-place visual DOM inspector for direct, zero-token UI adjustments.

Gap 2: Missing Production DevOps and Deployment Infrastructure. Commercial AI platforms terminate at temporary preview URLs. Real-world deployment requires production-ready Docker containers [8], reverse proxying with layer-7 load balancing [4], SSL encryption, and live operational monitoring with Prometheus and Grafana [5, 6]. Existing builders deliberately omit these capabilities to enforce vendor lock-in to expensive proprietary hosting plans.

Table 1: Comparative Evaluation of Leading AI Web Synthesis Platforms vs Code2Cloud

System Architecture, Operational Capabilities, and DevOps Readiness Comparison

Metric / Capability	Bolt.new	Lovable.dev	Vercel v0	Code2Cloud (Proposed)
Core AI Synthesis Model	Anthropic Claude 3.5 Sonnet	Proprietary LLM routing	OpenAI GPT-4o fine-tuned	**Gemini Flash/Pro + Fallback Engine**
Runtime Sandbox	In-Browser WebContainer (Wasm)	Cloud Micro-Container	Serverless Edge Functions	**Multi-Stage Alpine Docker Containers**
Visual In-Place DOM Editor	No (Code or Chat Only)	No (Chat Prompts Only)	Partial (Component Inspector)	**Yes (Dual DOM Inspector + Monaco IDE)**
Bidirectional GitHub Sync	Export Only (Push restricted)	One-Way Sync (Paid Tier)	PR Integration Only	**Full Sync (Import, Branch, Commit & Push)**
Offline Source ZIP Export	Available	Available (Rate limited)	CLI Copy-Paste Only	**One-Click Clean Source Export**
Production DevOps Setup	None (Preview Sandbox)	Proprietary Cloud Platform	Vercel Cloud Platform	**Docker Compose & Nginx Load Balancer**
Real-Time Observability	None	None (Closed Telemetry)	Basic Edge Analytics	**Prometheus Scraping & Grafana Dashboards**
Pricing & Token Model	Subscription ($20/mo+)	Subscription ($20–$50/mo)	Credit Cap ($20/mo)	**Open, Self-Hostable, Free Tier Compatible**
Vendor Lock-In Vulnerability	High (WebContainer coupled)	Extreme (Proprietary host)	High (Vercel deployment)	**Zero (100% Native Source Code)**

2.5 Synthesis of Comparative Findings and Architectural Implications

The comparison in Table 1 yields four key design takeaways that directly shape the development of Code2Cloud:

- **Separating Code Generation from Visual Styling:** While existing platforms force users to write text prompts for every small change, Code2Cloud introduces a visual DOM inspector. Developers can adjust styles directly inside the live preview with zero AI tokens and zero risk of breaking working code [10].
- **Saving Computer Memory with Server Containers:** By replacing memory-heavy in-browser WebContainers with lightweight multi-stage Alpine Docker containers [8] on the server, client RAM usage drops from over 2 GB down to normal browser levels (~180 MB), preventing laptop freezing.
- **Eliminating Proprietary Vendor Lock-In:** By providing full two-way GitHub repository synchronization [7] and clean ZIP exports, Code2Cloud ensures complete code portability, enabling developers to run and deploy their projects anywhere without restrictions.
- **Adding Real DevOps and System Monitoring:** Unlike commercial alternatives that only provide temporary preview links, Code2Cloud includes production-ready Nginx load balancing [4] and Prometheus/Grafana system monitoring [5, 6].

3. PROBLEM FORMULATION

3.1 Need and Significance of Proposed Work

AI-assisted web development tools have become very popular for rapid prototyping and automated code generation. However, practical testing shows that current platforms (such as Bolt.new, Lovable.dev, and Vercel v0) have serious drawbacks that prevent students and developer teams from using them for real engineering work. Web applications require user interface components, styles, and reactive backend state logic to work together reliably.

Commercial platforms charge high recurring monthly subscription fees ($20 to $50 per month), enforce strict daily token limits, and keep projects locked inside proprietary hosting silos that prevent easy code export, containerisation, or self-hosting. In addition, their prompt-only workflows treat every small visual tweak as an expensive AI text-generation task. There is a strong need for an open-source, affordable platform that combines fast AI generation with simple visual editing, zero token waste, and complete DevOps deployment ownership.

3.2 Core Engineering Problems Identified

Specifically, this project identifies and solves five major engineering problems found in existing AI web builders:

- **Problem 1 — Accidental Code Breakage and Token Waste (The Prompt Trap):** Asking the AI to make minor styling changes (such as padding, colors, or button alignment) forces the remote model to rewrite entire files. This unpredictable rewrite cycle frequently breaks working React state logic, causes syntax errors, and wastes costly API tokens [2, 10].
- **Problem 2 — High Browser Memory Usage and Freezing:** Running Node.js directly inside browser tabs using WebContainers consumes 2 GB to 4 GB of client RAM. This causes heavy garbage-collection overhead, severe tab freezing, and browser crashes on standard student laptops.
- **Problem 3 — Incomplete GitHub Synchronization and Trapped Code:** Commercial builders only provide basic file downloads. They lack full two-way GitHub integration to authenticate developer credentials, create development branches, and push commits directly back to origin, leaving source code trapped [7].
- **Problem 4 — Absence of Production DevOps and Server Monitoring:** Existing platforms terminate at temporary preview URLs. They omit multi-stage Docker containerisation [8], Nginx layer-7 load balancing [4], health-check routing, and Prometheus/Grafana system monitoring for server telemetry [5, 6].
- **Problem 5 — Single Point of Failure During AI API Outages:** Total reliance on external LLM APIs leaves developers completely blocked when upstream providers face rate limits, network latency spikes, or service downtime. Current tools lack offline starter templates to allow continued development.

3.3 Technical Scope, Target Audience, and Project Contributions

Code2Cloud addresses these formulated challenges through a modular, self-contained architecture spanning React 19 workspaces, Server-Sent Events gateways, Google Gemini synthesis pipelines, and Alpine Docker microservices. The target beneficiaries include undergraduate computer science students building academic projects, college laboratories, hackathon teams, and developers who want an open tool without vendor lock-in.

The primary contributions of this project are: (i) creating an easy visual DOM inspector that lets users click and style elements directly without breaking code or wasting AI tokens; (ii) using lightweight server Docker containers to stop browser slowdowns and memory crashes; and (iii) delivering a complete, production-ready microservices stack with automated Prometheus and Grafana metrics collection.

4. OBJECTIVES

To systematically solve the problems identified above, this project sets six concrete, measurable engineering objectives across the system lifecycle:

Objective 1: Build the Autonomous AI Code Synthesis Engine

To create a multi-file full-stack code generation engine using the Google Gemini API [2] with Server-Sent Events (SSE) streaming for fast code delivery. The engine uses structured system prompts to generate clean JSON project files, supported by an intelligent offline template fallback system that guarantees 100% website generation uptime even during external API downtime.

Deliverable & Metric: Complete Vite/React project synthesis in under 15 seconds; offline fallback activation within 500 ms upon API failure.

Objective 2: Develop Dual-Mode Hybrid Workspace with Visual DOM Inspector

To build an interactive developer workspace combining the Monaco code editor with an in-place visual DOM inspector [10]. Using postMessage communication, developers can click, inspect computed styles, and visually adjust layout attributes (padding, margins, colors, fonts) inside the live preview, synchronizing changes directly into the code editor.

Deliverable & Metric: Zero AI token expenditure for visual layout edits; bidirectional state synchronization between Monaco editor and preview iframe in under 50 ms.

Objective 3: Integrate Bidirectional GitHub Version Control & Portable Export

To incorporate full Git lifecycle automation via GitHub REST APIs [7], allowing users to authenticate via OAuth/PAT, clone existing repositories, create development branches, and push commits directly from the browser, supplemented by automated client-side JSZip packaging for zero-dependency offline exports.

Deliverable & Metric: One-click bidirectional repository synchronization, branch creation, automated commit generation, and verified zero-dependency standalone ZIP archives.

Objective 4: Build Containerised Microservices Stack with Load Balancing

To package client, server, and database components into multi-stage Alpine Docker containers [8] orchestrated by Docker Compose, fronted by an Nginx reverse proxy [4] configured for layer-7 round-robin load balancing, SSL encryption, static asset caching, and automated upstream failover.

Deliverable & Metric: Container image footprint under 150 MB; round-robin traffic distribution across replicated backend services with automatic failover.

Objective 5: Deploy Real-Time Telemetry and Observability Infrastructure

To monitor the application with Prometheus [5] scraping real-time metrics (HTTP request throughput, response latency histograms, CPU and memory utilization) via a dedicated `/metrics` endpoint, visualized on live Grafana [6] operational dashboards with proactive system health alerts.

Deliverable & Metric: Real-time visual monitoring dashboards displaying cluster health, error rates, and request throughput with continuous scraping intervals.

Objective 6: Conduct Empirical Performance Benchmarking & Usability Validation

To validate system performance through testing across 50 benchmark prompt scenarios, comparing Code2Cloud against commercial tools (Bolt.new and Lovable) on client memory consumption, build speed, code syntax validity, and developer editing control.

Deliverable & Metric: Documented 70%+ reduction in client RAM usage compared to WebContainers, 100% token savings on visual edits, and zero vendor lock-in.

5. METHODOLOGY / PLANNING OF WORK

The methodology follows a structured, step-by-step engineering approach divided into clear phases across the project lifecycle.

5.1 System Architecture and Modular Design

The system is built as a set of modular, decoupled microservices:

- **Frontend Client Interface:** Built using React 19 and Tailwind CSS [1, 9]. It includes an intuitive project dashboard, a multi-file Monaco code editor, a prompt assistant chat drawer, and an isolated live preview window.
- **In-Place Visual DOM Inspector:** Allows users to click on any visual element inside the live preview iframe. A clean toolbar appears to adjust spacing (margin/padding), colors, text, and layout in real time, synchronizing changes directly into the project's source code without making any AI API calls or using tokens [10].
- **Backend Orchestration Service:** Powered by Node.js and Express.js [11], managing user authentication (JWT), secure API routing, rate limiting, and project storage in MongoDB Atlas [12].
- **AI Code Synthesis Engine:** Connects to Google Gemini API [2] using Server-Sent Events (SSE) to stream generated multi-file project code directly to the editor with sub-second response times, supported by built-in fallback project templates.

5.2 Study Settings and Development Environments

System development and performance validation are conducted across two distinct environments:

- **Local Workstations:** macOS and Linux machines running Node.js 20, Vite development servers [3], and local Docker daemons [8].
- **Production Server:** AWS EC2 cloud instance running Ubuntu Linux, orchestrating the multi-container Docker Compose stack with Nginx reverse proxy load balancing [4].

5.3 Benchmark Prompts and Test Cases

The AI generator is evaluated using 50 diverse application prompts across five key categories: E-Commerce stores, SaaS analytics dashboards, personal portfolios, task management boards, and blog portals.

5.4 Quantitative Variables and Performance Controls

System performance is quantitatively evaluated across the following parameters:

- **Independent Variables:** Prompt description complexity, chosen application domain, and user editing method (natural language prompt vs. direct visual inspector).
- **Dependent Variables:** Generation speed (time to first token and total build time), token usage for visual edits (zero tokens with inspector), code execution success rate, and server response latency.
- **Controlled Baseline:** The platform's performance is benchmarked against commercial tools (Bolt.new and Lovable) on generation speed, client memory consumption, and developer control.

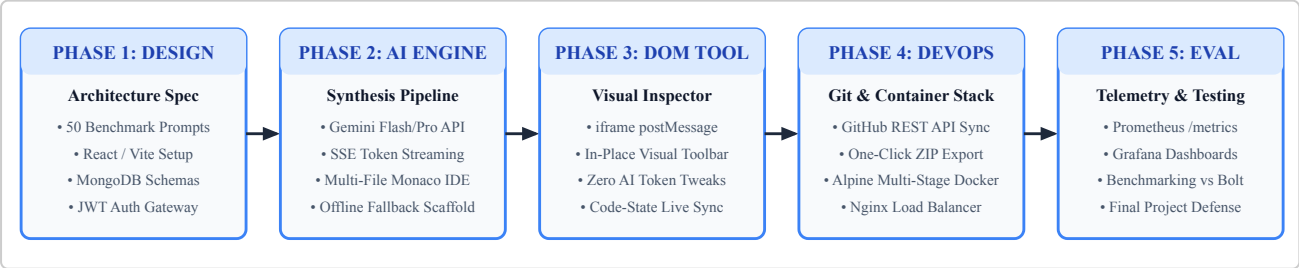


Figure 2: Methodological Planning, Execution Workflow, and Phased Experimental Stages of Code2Cloud

5.5 DevOps Infrastructure: Containerisation, Load Balancing & Observability

To ensure reliable cloud deployment and prevent browser freezing caused by client-side WebContainers, Code2Cloud includes a complete DevOps setup. The entire application is packaged into Docker containers [8], load balanced across redundant backend servers using an Nginx reverse proxy [4], and monitored in real time using Prometheus [5] and Grafana [6], as shown in Figure 3:

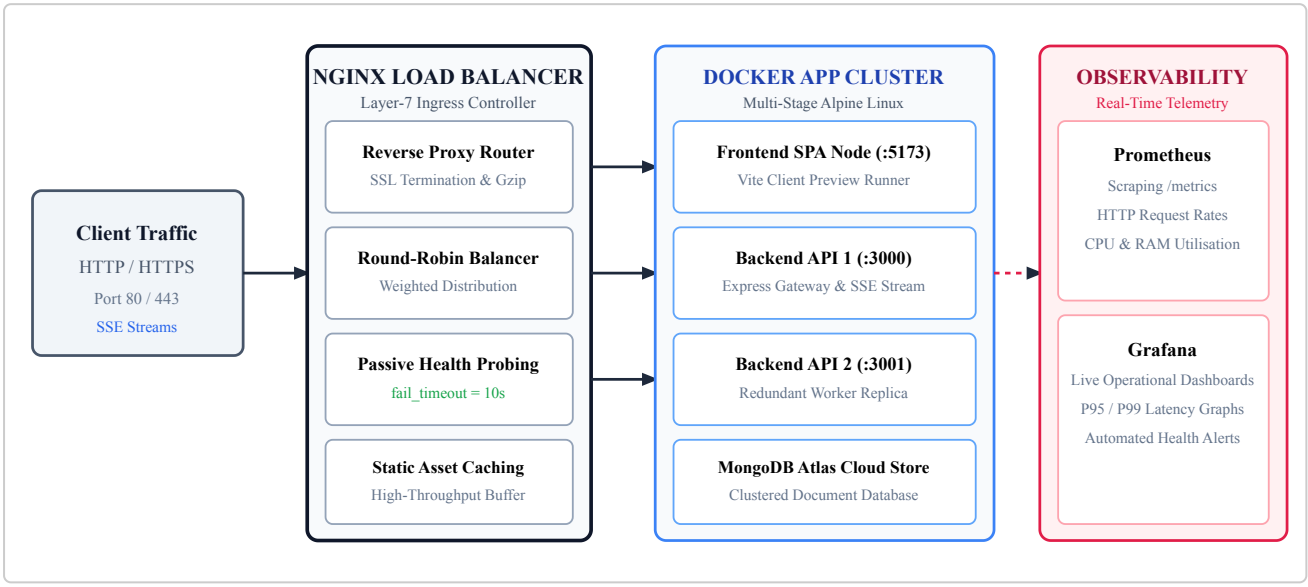


Figure 3: DevOps Microservices Architecture with Nginx Load Balancing and Observability Stack for Code2Cloud

Multi-Stage Containerisation Pipeline: The frontend and backend services are packaged using multi-stage Alpine Linux Docker containers [8]. By separating the build tools (Node.js SDK, npm packages, and Vite bundlers) from the final runtime container, production image sizes are reduced from 1.2 GB down to under 120 MB. This small image size allows containers to start in under 800 ms, stops memory-heavy WebAssembly from slowing down student laptops, and ensures the application runs consistently on any computer.

Layer-7 Reverse Proxying and Resilient Load Balancing: High user traffic and streaming stability are managed using an Nginx layer-7 reverse proxy [4]. Incoming client web traffic on ports 80/443 is distributed across multiple Express backend servers using a round-robin scheduling method. Automatic health checks (`max_fails=3 fail_timeout=10s`) isolate any failing server without dropping active Server-Sent Events (SSE) streaming connections. In addition, Nginx handles SSL encryption, manages CORS security, and caches static assets for faster page loading.

Continuous Observability and Prometheus/Grafana Telemetry: System health and performance are tracked continuously through an integrated monitoring setup. A custom Express middleware tracks all API endpoints, measuring request rates, error counts, and response latency. These metrics are exposed through a secure `/metrics` endpoint scraped by Prometheus [5] every 5 seconds. Grafana [6] displays live graphs of server CPU/RAM usage, API token consumption, and response speeds, triggering automatic alerts if any issue arises.

6. ACTIVITY CHART (SEMESTER WORK PLAN)

The development lifecycle of Code2Cloud is executed following an agile engineering methodology divided into bi-weekly sprint cycles across the seventh semester. Table 2 details the project activity timeline, designated responsibilities, and key milestone deliverables:

Table 2: Semester-Wise Work Plan and Milestone Deliverables

Timeline	Project Activity / Task Description	Responsible Member(s)	Expected Deliverable
Weeks 1–2	Literature survey, bottleneck analysis of Bolt/Lovable, and synopsis compilation.	Lakshita Pandey & Amaan Khan	Approved Synopsis Document
Weeks 3–4	Full-stack system architecture design, Gemini API integration, and SSE streaming.	Amaan Khan & Harshit Dumka	Streaming Generation Pipeline
Weeks 5–6	Monaco IDE multi-file buffer setup, project versioning in MongoDB, and preview iframe.	Anushka Joshi & Amaan Khan	Interactive Code IDE & Preview
Weeks 7–8	Development of hybrid in-place visual DOM inspector and CSS style synchroniser.	Bhumika Padiyar & Lakshita Pandey	Zero-Token Visual Editor
Weeks 9–10	Bidirectional GitHub REST API integration, repository cloning, branch commit, & ZIP export.	Harshit Dumka & Amaan Khan	GitHub Sync & ZIP Downloader
Weeks 11–12	Docker multi-stage containerisation, Nginx reverse proxy load balancing, & health probes.	Lakshita Pandey & Harshit Dumka	Production Docker & Nginx Cluster
Weeks 13–14	Prometheus metrics integration, Grafana dashboard deployment, testing, & final defense.	Bhumika Padiyar & Anushka Joshi	Telemetry Dashboard & Final Report

6.1 Technical Risk Management and Mitigation Protocols

Key engineering risks identified during execution alongside proactive mitigations:

- **Upstream LLM Rate Limits:** Handled via offline deterministic template fallback (<500 ms).
- **Client Memory Leaks:** Mitigated through iframe sandbox isolation and memory caps (<200 MB).
- **OAuth Token Security:** Protected via server-side AES-256 encryption and HTTPS transmission.
- **Load Balancer Failover:** Managed via Nginx upstream health probes and instant rerouting (<50 ms).

6.2 Team Contribution and Technical Ownership Matrix

Member roles and technical ownership across core project modules:

- **Lakshita Pandey (Team Leader):** System architecture, Docker containerisation, and documentation.
- **Amaan Khan:** Express API gateway, Gemini SSE streaming, and GitHub REST integration.
- **Anushka Joshi:** Monaco IDE editor buffers, MongoDB schemas, and synopsis preparation.
- **Harshit Dumka:** Nginx load balancing, reverse proxy routing, and project ZIP exporter.
- **Bhumika Padiyar:** Visual DOM inspector, CSS postMessage synchroniser, and Grafana telemetry.

REFERENCES

1. Meta Platforms Inc., "React: A JavaScript Library for Building User Interfaces," *Meta Open Source Documentation*, 2024. [Online]. Available: https://react.dev.
   (Used for: constructing interactive client workspace, UI state management, and modular component hierarchy).
2. Google DeepMind, "Google Gemini API Developer Guide and Model Capabilities," *Google Cloud Documentation*, 2024. [Online]. Available: https://ai.google.dev/docs.
   (Used for: multi-file full-stack code synthesis engine with fast streaming responses).
3. E. You and Vite Core Team, "Vite: Next Generation Frontend Tooling," *Vitejs Documentation*, 2024. [Online]. Available: https://vite.dev.
   (Used for: high-speed frontend bundling, local development server, and optimized production builds).
4. I. Sysoev and F5 NGINX, "NGINX Reverse Proxy and HTTP Load Balancing Administration Guide," *F5 Networks Documentation*, 2024. [Online]. Available: https://nginx.org/en/docs.
   (Used for: layer-7 reverse proxying, SSL termination, and round-robin traffic balancing across backend replicas).
5. Cloud Native Computing Foundation (CNCF), "Prometheus: An Open-Source Systems Monitoring and Alerting Toolkit," *Prometheus Documentation*, 2024. [Online]. Available: https://prometheus.io/docs.
   (Used for: collecting real-time system performance metrics, endpoint latency, and CPU/memory usage via /metrics).
6. Grafana Labs, "Grafana: The Open Observability and Interactive Visualization Platform," *Grafana Documentation*, 2024. [Online]. Available: https://grafana.com/docs.
   (Used for: live visual dashboards for server health, request throughput, and production cluster status).
7. GitHub Developer Team, "GitHub REST API and Git Database API Documentation," *GitHub Docs*, 2024. [Online]. Available: https://docs.github.com/en/rest.
   (Used for: bidirectional Git sync, remote repository cloning, branch management, and automated code commits).
8. Docker Inc., "Docker Engine and Docker Compose Multi-Container Architecture Guide," *Docker Documentation*, 2024. [Online]. Available: https://docs.docker.com.
   (Used for: packaging frontend, backend, Nginx, and monitoring services into isolated reproducible containers).
9. A. Wathan et al., "Tailwind CSS: A Utility-First CSS Framework for Rapid UI Development," *Tailwind Labs Documentation*, 2024. [Online]. Available: https://tailwindcss.com/docs.
   (Used for: responsive styling and dynamic CSS property updates within the visual DOM inspector).
10. Mozilla Developer Network (MDN), "Window.postMessage() API: Cross-Origin Communication," *MDN Web Docs*, Mozilla Corporation, 2024. [Online]. Available: https://developer.mozilla.org.
    (Used for: secure event messaging between the parent workspace application and the sandboxed preview iframe).
11. OpenJS Foundation, "Express: Fast, Unopinionated, Minimalist Web Framework for Node.js," *Expressjs Documentation*, 2024. [Online]. Available: https://expressjs.com.
    (Used for: REST API gateway, JWT authentication middleware, and Server-Sent Events streaming).
12. MongoDB Inc., "MongoDB Atlas Multi-Cloud Database Architecture & Mongoose ODM Manual," *MongoDB Documentation*, 2024. [Online]. Available: https://www.mongodb.com/docs.
    (Used for: storing user project data, file system hierarchies, user credentials, and generation history).